

เอกสารทบทวนสัปดาห์ที่ 12

เรื่อง การหาผลเฉลยในรูปอนุกรมกำลังของสมการเชิงอนุพันธ์

1. จงหาผลเฉลยในรูปอนุกรมกำลังของสมการเชิงอนุพันธ์

$$y' - 2xy = 0$$

วิธีทำ ให้ผลเฉลยในรูป

$$y = \sum_{n=0}^{\infty} a_n x^n$$

หาอนุพันธ์ได้

$$y' = \sum_{n=0}^{\infty} a_n (n x^{n-1}) = 0 + \sum_{n=1}^{\infty} n a_n x^{n-1}$$

แทนในสมการ $y' - 2xy = 0$

ได้ว่า

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$2xy = 2x \left(\sum_{n=0}^{\infty} a_n x^n \right)$$

$$2xy = \sum_{n=0}^{\infty} 2a_n x^{n+1}$$

$$\sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} 2a_n x^{n+1} = 0$$

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n - \sum_{n=1}^{\infty} 2a_{n-1} x^n = 0$$

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n - \sum_{n=1}^{\infty} 2 a_{n-1} x^n = 0$$

$$a_1 + \sum_{n=1}^{\infty} (n+1) a_{n+1} x^n - \sum_{n=1}^{\infty} 2 a_{n-1} x^n = 0$$

$$a_1 + \sum_{n=1}^{\infty} x^n ((n+1) a_{n+1} - 2 a_{n-1}) = 0$$

$$a_1 = 0$$

$$(n+1) a_{n+1} - 2 a_{n-1} = 0$$

↓

$$(n+1) a_{n+1} = 2 a_{n-1}$$

$$a_{n+1} = \frac{2 a_{n-1}}{n+1}$$

จากการเทียบสัมประสิทธิ์ พบว่า $a_1 = 0$

และ $a_{n+1} = \frac{2a_{n-1}}{n+1}, n=1,2,3,\dots$

เมื่อ $n=1$: $a_2 = \frac{2a_0}{2} = a_0$

$n=2$: $a_3 = \frac{2a_1}{3} = 0$

$n=3$: $a_4 = \frac{2a_2}{4} = \frac{a_2}{2} = \frac{a_0}{2}$

$n=4$: $a_5 = \frac{2a_3}{5} = 0$

$n=5$: $a_6 = \frac{2a_4}{6} = \frac{a_4}{3} = \frac{a_0}{6}$

⋮

$$y = a_0 + 0x + \frac{a_0}{\sqrt{2}}x^2 + 0x^3 + \frac{a_0}{6}x^4 + \dots$$

$$y = a_0 + \frac{a_0}{\sqrt{2}}x^2 + \frac{a_0}{6}x^4 + \frac{a_0}{4!}x^6 + \dots$$

$$y = a_0 \left(1 + \frac{x^2}{\sqrt{2}} + \frac{x^4}{6} + \frac{x^6}{4!} + \dots \right)$$

2. จงหาผลเฉลยในรูปอนุกรมกำลังของสมการเชิงอนุพันธ์

$$y'' + xy' + 2y = 0$$

สมมติให้ $y = \sum_{n=0}^{\infty} a_n x^n$

$$y' = \sum_{n=0}^{\infty} a_n (n x^{n-1}) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=1}^{\infty} n a_n ((n-1) x^{n-2})$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

$$x y' = x \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^n$$

$$2y = 2 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} 2a_n x^n$$

$$0 + \sum_{n=1}^{\infty} n a_n x^n = 0 \cdot a_0 x^0 + \sum_{n=1}^{\infty} n a_n x^n = \sum_{n=0}^{\infty} n a_n x^n$$

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} 2 a_n x^n = 0$$

$$\sum_{n=0}^{\infty} x^n [(n+2)(n+1) a_{n+2} + (n+2) a_n] = 0$$

$$(n+2)(n+1) a_{n+2} + (n+2) a_n = 0$$

$$(n+1) a_{n+2} + a_n = 0$$

$$a_{n+2} = \frac{-a_n}{n+1} \quad ; n = 0, 1, 2, \dots$$

$$n=0; \quad a_2 = \frac{-a_0}{1} = -a_0$$

$$n=1; \quad a_3 = \frac{-a_1}{2}$$

$$n=2; \quad a_4 = \frac{-a_2}{3} = \frac{-(-a_0)}{3} = \frac{a_0}{3}$$

$$n=3; \quad a_5 = \frac{-a_3}{4} = \frac{-(-a_1/2)}{4} = \frac{a_1}{8}$$

↓;

$$y = a_0 + a_1 x - a_0 x^2 - \frac{a_1}{2} x^3 + \frac{a_0}{3} x^4 + \frac{a_1}{8} x^5 + \dots$$

3. จงหาผลเฉลยในรูปอนุกรมกำลังของสมการเชิงอนุพันธ์

$$\textcolor{red}{A} \quad \textcolor{red}{(x^2 + 3)y''} - \textcolor{red}{7xy'} + \textcolor{red}{16y} = 0, \quad y(0) = 3, y'(0) = 0$$

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y' = \sum_{n=0}^{\infty} n a_n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=1}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$A = (x^2 + 3) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

$$A = \underbrace{\sum_{n=2}^{\infty} n(n-1) a_n x^n}_{\textcolor{red}{A_1}} + \underbrace{3 \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}}_{\textcolor{red}{A_2}}$$

$$B = 7x \sum_{n=1}^{\infty} n a_n x^{n-1} = \underbrace{\sum_{n=1}^{\infty} 7n a_n x^n}_{\textcolor{red}{B_1}}$$

$$C = \underbrace{\sum_{n=0}^{\infty} 16 a_n x^n}_{\textcolor{red}{C_1}}$$

$$A_1 = \sum_{n=1}^{\infty} n(n-1) a_n x^n = \sum_{n=0}^{\infty} n(n-1) a_n x^n$$

$$B = \sum_{n=1}^{\infty} 7n a_n x^n = \sum_{n=0}^{\infty} 7n a_n x^n$$

$$C = \sum_{n=0}^{\infty} 16 a_n x^n$$

$$A_2 = \sum_{n=1}^{\infty} 3n(n-1) a_n x^{n-1}$$

$$A_2 = \sum_{n=0}^{\infty} 3(n+2)(n+1) a_{n+2} x^n$$

$$A_1 + A_2 - B + C = 0$$

$$\sum_{n=0}^{\infty} x^n [n(n-1) a_n - 7n a_n + 16 a_n + 3(n+2)(n+1) a_{n+2}] = 0$$

$$a_n (n(n-1) - 7n + 16) + 3(n+2)(n+1) a_{n+2} = 0$$

$$a_n (n^2 - 8n + 16) + 3(n+2)(n+1) a_{n+2} = 0$$

$$a_n (n-4)^2 + 3(n+2)(n+1)a_{n+2} = 0$$

$$a_{n+2} = \frac{-a_n (n-4)^2}{3(n+2)(n+1)} ; n = 0, 1, 2, \dots$$

$$y(0) = 3 \rightarrow a_0 = 3$$

$$y'(0) = 0 \rightarrow a_1 = 0$$

$$n=0 ; a_2 = \frac{-a_0(16)}{3(2)(1)} = -8$$

$$n=1 ; a_3 = -a_1(\dots) = 0$$

$$n=2 ; a_4 = \frac{-a_2(4)}{3(4)} = \frac{-(-8)}{12} = \frac{2}{3}$$

$$n=3 ; a_5 = -a_3(\dots) = 0$$

$$n=1, 3, 5, 7, \dots \rightarrow a_1, a_3, a_5, \dots = 0$$

$$n=4 ; a_6 = -a_4(0) = 0$$

$$n=6 \text{ ; } a_8 = -a_6 (\dots) = 0$$

$$\underline{a=6}$$

$$y = 3 - 8x^2 + \frac{8}{9}x^4$$